

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11: EXCRETION AND HOMEOSTASIS IN ANIMALS

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Welcome to your Final Explanation. Think back to the hot afternoon in a Kenyan community: a dog pants with its tongue out, a lizard sits still on a warm rock, goats drink water yet make concentrated urine, and freshwater fish keep excreting large amounts of dilute urine. Use everything you observed and learned in Lessons 1 to 7 to answer the questions below. Write your answers in the space under each question. Use evidence from your kidney dissection, your charts, your 'Follow the Fluid' table, your feedback flowcharts and your class model. Use correct terms such as nitrogenous waste, ultrafiltration, reabsorption, ADH, negative feedback, endotherm and ectotherm. Where a question asks for a claim, evidence and reasoning, state your claim first, support it with evidence, then explain why the evidence supports it. Your last task is to plan the annotated poster that will be your final product.

Part 1: Getting rid of nitrogenous wastes in different habitats

Animals make toxic nitrogenous wastes when they break down proteins. (a) Name the three nitrogenous wastes you studied and say which animals you linked to each one: freshwater fish, marine fish, insects, birds and mammals. (b) Explain how the type of waste an animal makes is linked to how much water is available to it. Use the freshwater fish and the insects and birds in your explanation. (c) Name the excretory structures used by freshwater fish, marine fish, insects and birds.

Write your answer here:

Part 2: How the mammalian kidney makes urine

Think about the goat kidney you dissected and the animations of the nephron. (a) Describe what you saw in the cortex, medulla and pelvis, and say what each region does. (b) Using your 'Follow the Fluid' table, explain the steps by which blood becomes urine. Include ultrafiltration, selective reabsorption and secretion. (c) Explain why glucose is found in blood and filtrate but not in urine, while urea is more concentrated in urine. (d) Which hormone controls how much water is reabsorbed, and where in the nephron does it act?

Write your answer here:

Part 3: The liver and skin, and the goat and the freshwater fish

(a) Where does the urea in a goat's urine come from, and how does it reach the urine? (b) State one other job of the liver and one job of the skin in excretion. (c) Explain why a goat that has lost water on a hot day makes small amounts of concentrated urine, while a freshwater fish makes large amounts of dilute urine. Write this as a claim, evidence and reasoning explanation, using what you learned about the kidney for the goat and about water movement for the fish.

Write your answer here:

Part 4: Negative feedback and the dog and the lizard

(a) Explain what negative feedback means, using a set point. (b) Draw or describe a feedback loop for either blood glucose after a meal of ugali and sukuma wiki, or body temperature on a hot day. Name the detector, the control centre and the effectors or hormones. (c) Explain why the dog pants and why the lizard sits still on the warm rock. In your answer, use the words endotherm and ectotherm. (d) Name two behaviours, other than basking and seeking shade, that animals use to survive extreme cold or dry conditions.

Write your answer here:

Part 5: Your final poster and why homeostasis matters

Your final product is an annotated poster model explaining how a mammal (a goat or a human) regulates its internal environment and excretes waste. (a) List at least five labelled parts or processes your poster must include and say what each shows. (b) Explain how your poster model is different from your first model in Lesson 1. (c) Explain why excretion and homeostasis are essential for the survival of the cells and health of all living organisms.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining excretion and nitrogenous wastes	Correctly names ammonia, urea and uric acid, matches them to animals and links each to water availability with clear reasoning.	Names the wastes and matches most to the correct animals, with a basic link to water availability.	Names some wastes but mixes up animals or does not link waste type to water.
Kidney structure and urine formation	Accurately describes cortex, medulla and pelvis and explains ultrafiltration, selective reabsorption, secretion and the role of ADH in the correct order, using evidence from the Follow the Fluid table.	Describes kidney regions and most urine formation steps, with minor gaps in sequence or terms.	Names some parts or steps but with inaccuracies or missing links between structure and function.
Negative feedback and thermoregulation	Explains set point, detector, control centre and effectors correctly for at least one loop, and clearly contrasts the dog (endotherm) and lizard (ectotherm).	Gives a mostly correct feedback loop and a reasonable explanation of the dog and lizard.	Describes a response but does not show the feedback idea or confuses endotherm and ectotherm.
Use of evidence and claim-evidence-reasoning	Makes clear claims supported by specific evidence from the lessons and reasoning that connects the evidence to the hot afternoon phenomenon.	Makes claims with some evidence, but the reasoning is partly incomplete.	Gives statements with little or no evidence or reasoning.
Poster model and communication	Poster plan includes at least five accurate, labelled parts with functions, shows links between kidney, liver, skin and feedback control, and explains why homeostasis matters for cells.	Poster plan includes about four accurate labelled parts with some links between them.	Poster plan has few or unclear labels and little connection between parts.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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